

Advanced Lesson 14 - Exercises:

Dynamic Interaction, Files and Strings

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MATH 7013 - Advanced Mathematical Modeling

1: Manipulate - dynamic 2-D and 3-D plots

Exercises 1 and 2 repeat Lesson 4's Exercises 14 and 15 verbatim. The cells below build correctly under a plain kernel, but the sliders are live only inside a Wolfram front end. Evaluated through Wolfram Engine under Jupyter they render as a static snapshot at the initial parameter values, so these cells are left unevaluated here.

```
Manipulate[Plot[E^(-a x), {x, 0, 1}], {a, 0, 10}]
```

```
Manipulate[Plot[E^(-a x), {x, 0, 1}, PlotRange -> All], {a, 0, 10}]
```

```
Manipulate[  
  Plot3D[E^(-a x - b y), {x, 0, 1}, {y, 0, 1}, PlotRange -> All],  
  {a, 0, 10}, {b, 0, 10, 2}]
```

2: A dynamic algebraic calculator

```
Manipulate[Expand[(x + y)^n], {n, 0, 10, 1}]
```

3: Writing a list to disk, reading it back

```
Clear[dataList, myData, file];
dataList = {1, 2, 3, 4, 5, 6, 7};
file = FileNameJoin[{$TemporaryDirectory, "dataList.txt"}];
Export[file, dataList, "List"];
file

FilePrint[file]

myData = Import[file, "List"];
myData

{Head[myData], Length[myData], myData === dataList}
```

4: Turning text into a list of characters

```
Characters["Mathematica"]
Length[Characters["Mathematica"]]
```